

WHEEL TYPE ROTARY ENCODERS

Wheel type Incremental rotary encoder (EW Series)

EW-E-01-003002

※ It is an encoder of reliable and excellent quality that is applicable to the automated control industry.
 ※ The above specifications are subject to change without notice.

Ordering Information

EW	1mm	T	24
Series	Min. measuring unit (1Pulse)	Control output	Power supply
EW: Incremental	1mm, 2mm, 2.5mm, 5mm 1cm, 2cm, 2.5cm, 5cm, 10cm 1m, 0.01yd, 0.1yd, 1yd	T: Totem pole output N: NPN NPN O.C output V: Voltage output L: Line driver output	5 : 5VDC 24: 12~24VDC

Connections

■ T: Totem pole Output N: NPN O.C Output V: Voltage Output

Function	Color	Function	Color
OUT A	Green	+V	Red
OUT B	White	GND(0V)	Black
OUT Z	Blue	Shield	F.G.

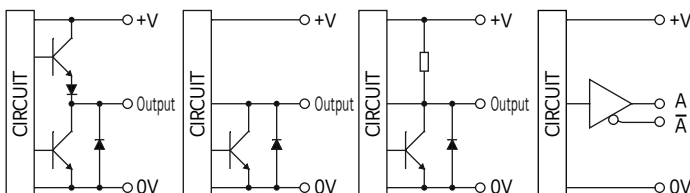
■ L: Line driver Output

Function	Color	Function	Color
OUT A	Green	OUT $\bar{A}$	Brown
OUT B	White	OUT $\bar{B}$	Orange
OUT Z	Blue	OUT $\bar{Z}$	Yellow
+V	Red	Shield	F.G.
GND(0V)	Black	-	

※ Unused wires must be insulated.
 ※ The encoder and shield cable must be grounded (F.G.).
 ※ F.G. (Frame Ground) must be grounded separately.

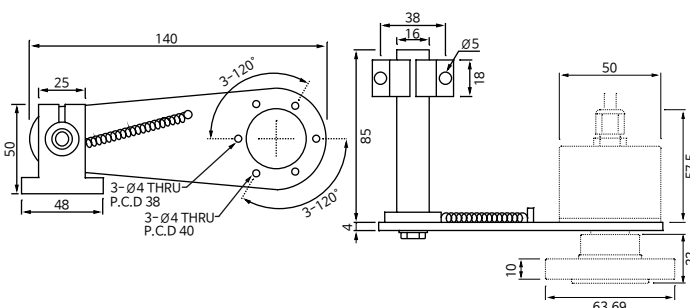
Control output diagram

■ Totem pole Output ■ NPN O.C Output ■ Voltage Output ■ Line driver Output



Dimensions

※Unit:mm



※ Wheel circumference : 200mm

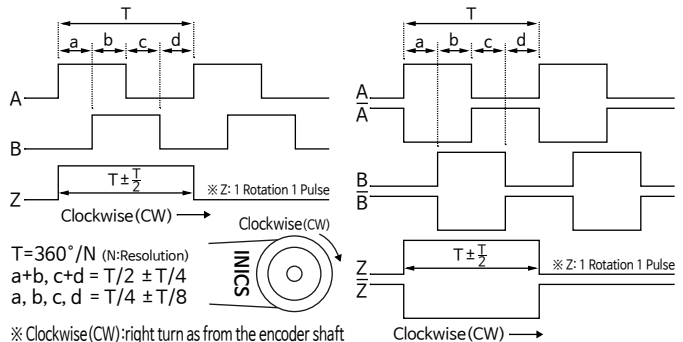
Specifications

Model		Wheel type Incremental rotary encoder
Power supply		5VDC±5%, 5VDC~24VDC±5% (Ripple P-P : Max. 5%)
Current consumption		Max. 100mA(no load)
Resolution(P/R)		1mm, 2mm, 2.5mm, 5mm, 1cm, 2cm, 2.5cm, 5cm, 10cm, 1m, 0.01yd, 0.1yd, 1yd
Control Output	Totem pole output	「L」Level – Load current: Max.10mA, Residual voltage: Max. 0.5VDC 「H」Level – Load current: Max.10mA Power voltage: Min. (Power voltage–2.5VDC)
	NPN O.C output	Load current: Max. 20mA, Residual voltage: Max. 0.5VDC
	Voltage output	Load current: Max. 10mA, Residual voltage: Max. 0.5VDC
	Line driver output	Load current : 「H」Level I _o = Max. –20mA, 「L」Level I _s = Max. 20mA Output voltage: Min. (Power voltage–2.5VDC), V _s = Max. 0.5VDC
Response time	Totem pole output	Max. 1μs (Cable length : 1.5m, I sink=20mA)
	NPN O.C output	Max. 2μs (Cable length : 1.5m, I sink=20mA)
	Voltage output	
	Line driver output	Max. 1μs (Cable length : 1.5m, I sink=20mA)
Max. Response frequency		10kHz
Max. Allowable revolution		5,000rpm
Shaft loading		Radial: Max. 2kg·f, Thrust: Max. 1kg·f
Shock		Max. 50G
Vibration		Max. 5G
Protection		IP50 (IEC standard)
Environment	Ambient temperature	–10 ~ 70℃ (no freezing or condensation)
	Storage temperature	–20 ~ 85℃ (no freezing or condensation)
	Ambient humidity	35% ~ 85% RH
	Storage humidity	35% ~ 90% RH
Cable		Shield cable, Length: 1.5m
Accessory		Product, Bracket, Bolt, Instruction manual
Approval		CE※excluding some items

※ Select resolution to satisfy Max. allowable revolution $\geq$ Max. response revolution
 ※ Max. response revolution (rpm) = (Max. response frequency / resolution) X 60 sec

Output Waveform

■ Totem pole Output, NPN O.C Output, Voltage Output ■ Line driver Output



※ Clockwise (CW): right turn as from the encoder shaft

Cautions during use

Environment

- Please do not use this unit with below environment, or it may cause malfunction.
- 1) Place where this unit or component may be damaged by strong vibration or impact.
- 2) Place where there is a lot of flammable or corrosive gases.
- 3) Place where strong magnet field or electric noise occurs.
- 4) Place where strong acids or alkali near and direct ray of the sun.

Cautions during installation

- 1) Do not load overweight on the shaft.
- 2) Do not put strong impact when insert a coupling into shaft.
- 3) If the coupling error (parallel misalignment, angular misalignment) between the shaft increases while installation, the life cycle of the coupling and the encoder can be shorten.
- 4) Do not connect, repair, or inspect the unit while connected to a power source.
- 5) If a big impact or strong vibration applies to the product it may cause pulse errors.

Wire Connection

- 1) Do not pull out the wire with over 20N strength after fixing the unit and wiring the cable.
- 2) if wire encoder cable with high voltage line or power cable in the same conduit, it may cause a malfunction or mechanical problem. Please wire it separately or use separated conduit.

Fail-safe device must be installed when using the unit with machinery that may cause serious injury or substantial economic loss.

- 1) medical equipment, ship, vehicle, railways, aircraft, safety equipment, etc
- 2) Failure to follow this instruction may result in personal injury, economic loss or fire